Embedded Control for Robotics

Lecture # 04 **Laplace Transform & Pole/Zero Stability**

- Book: Classical and Modern Control with Microcontrollers, Ch 2, 4
- Control Systems Engineering - Design and Implementation using Arm Cortex-M Microcontrollers: Ch 3

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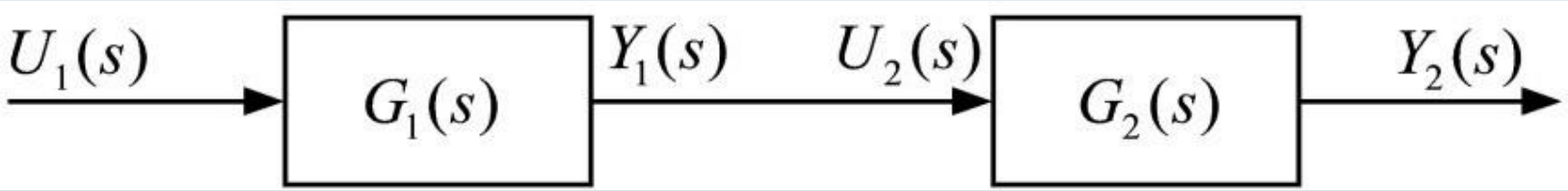
Overview Today's Lecture

- Block Diagram Algebra
- Laplace Transform
- Transfer Function
- Pole/Zero and Stability
- Python Demo

Block Diagram Algebra

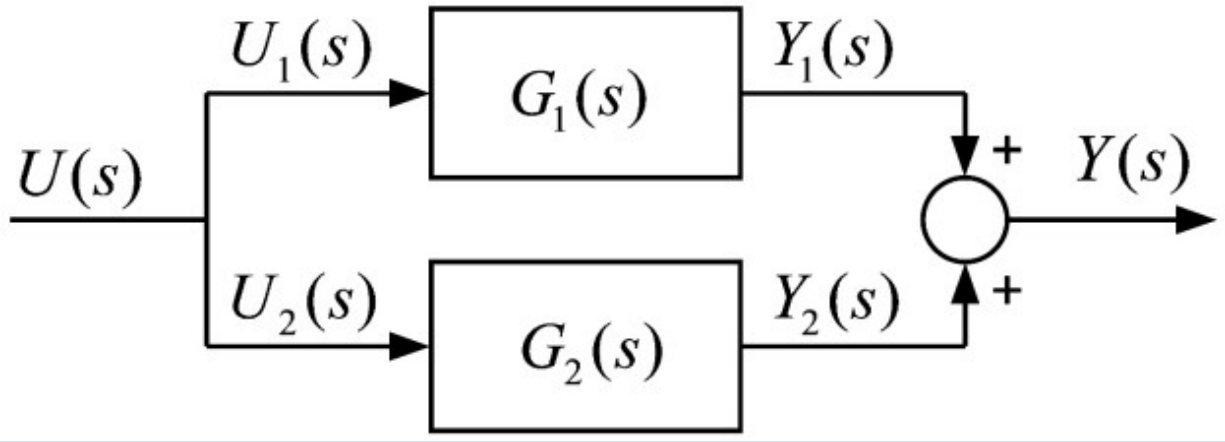
- Simplification rules
  - Series
  - Parallel
  - Feedback

Block Diagram Algebra: Series



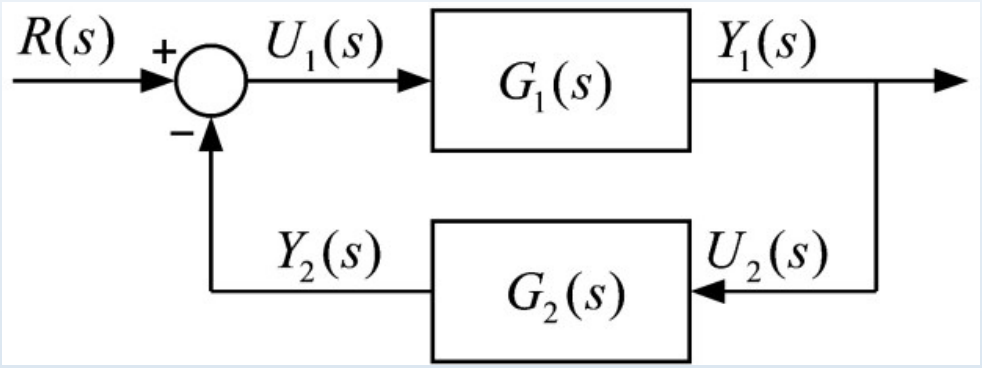
$$G(s) = \frac{Y_2(s)}{U_1(s)} = \frac{Y_2(s)}{U_2(s)} \frac{Y_1(s)}{U_1(s)} = G_2(s)G_1(s)$$

Block Diagram Algebra: Parallel



$$G(s) = \frac{Y(s)}{U(s)} = \frac{Y_1(s) + Y_2(s)}{U(s)} = \frac{Y_1(s)}{U_1(s)} + \frac{Y_2(s)}{U_2(s)} = G_1(s) + G_2(s)$$

Block Diagram Algebra: Feedback

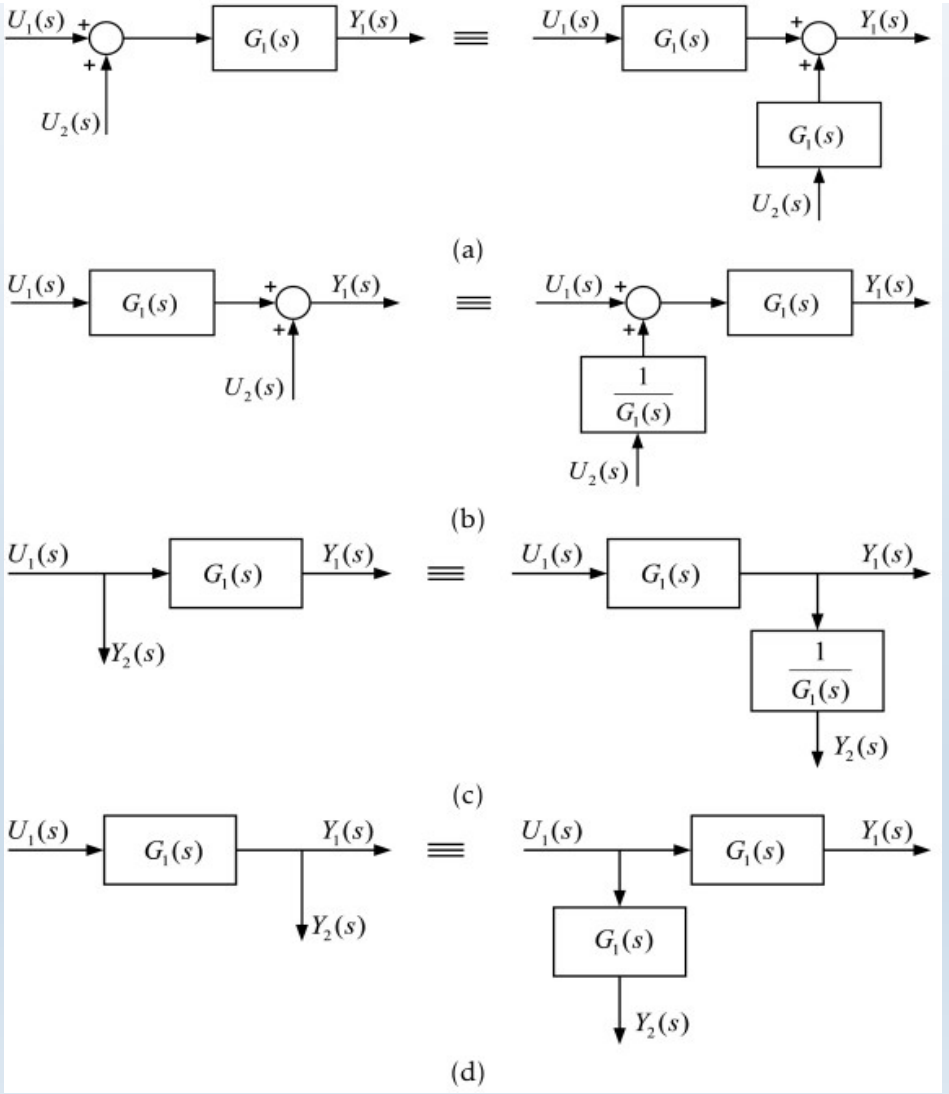


$$U_1(s) = R(s) - Y_2(s) = R(s) - G_2(s)Y_1(s)$$

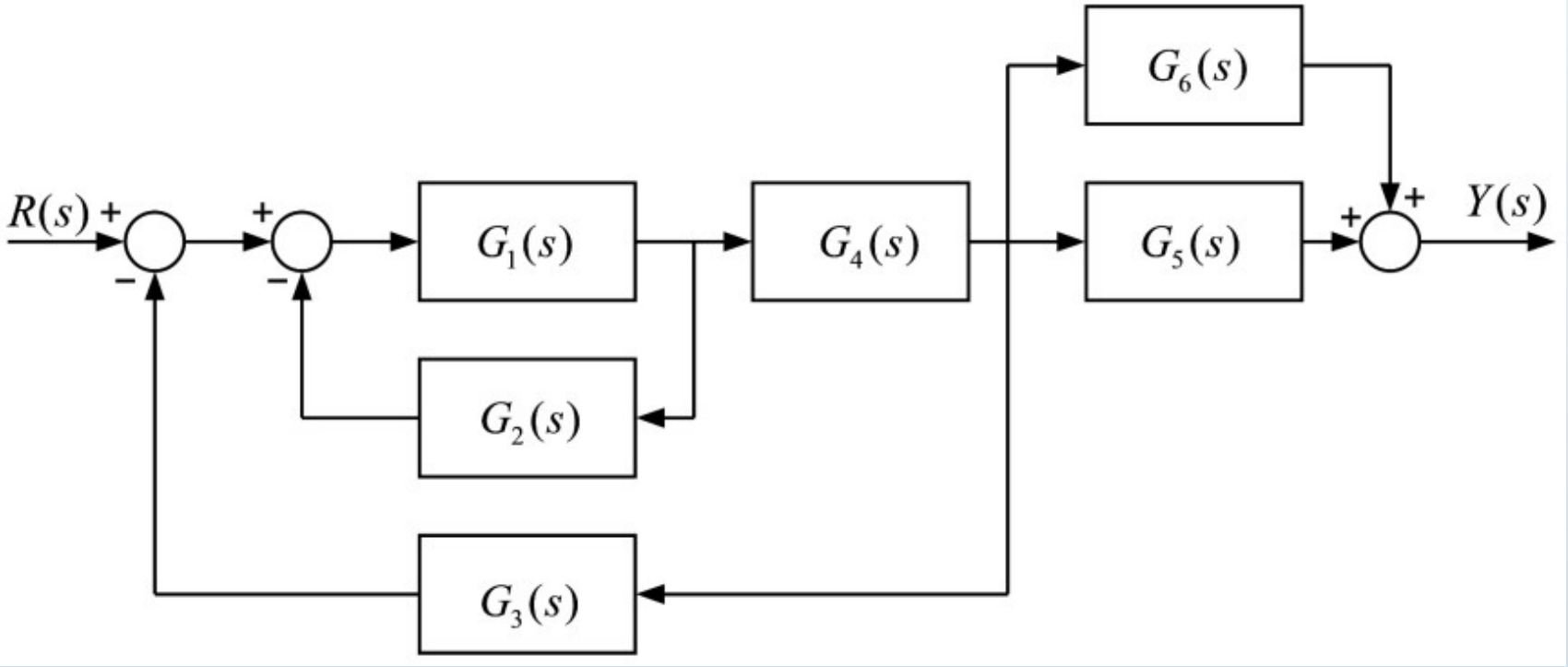
$$Y_1(s) = G_1(s)U_1(s) = G_1(s)\left(R(s) - G_2(s)Y_1(s)\right)$$

$$G(s) = \frac{Y_1(s)}{R(s)} = \frac{G_1(s)}{1 + G_1(s)G_2(s)}$$

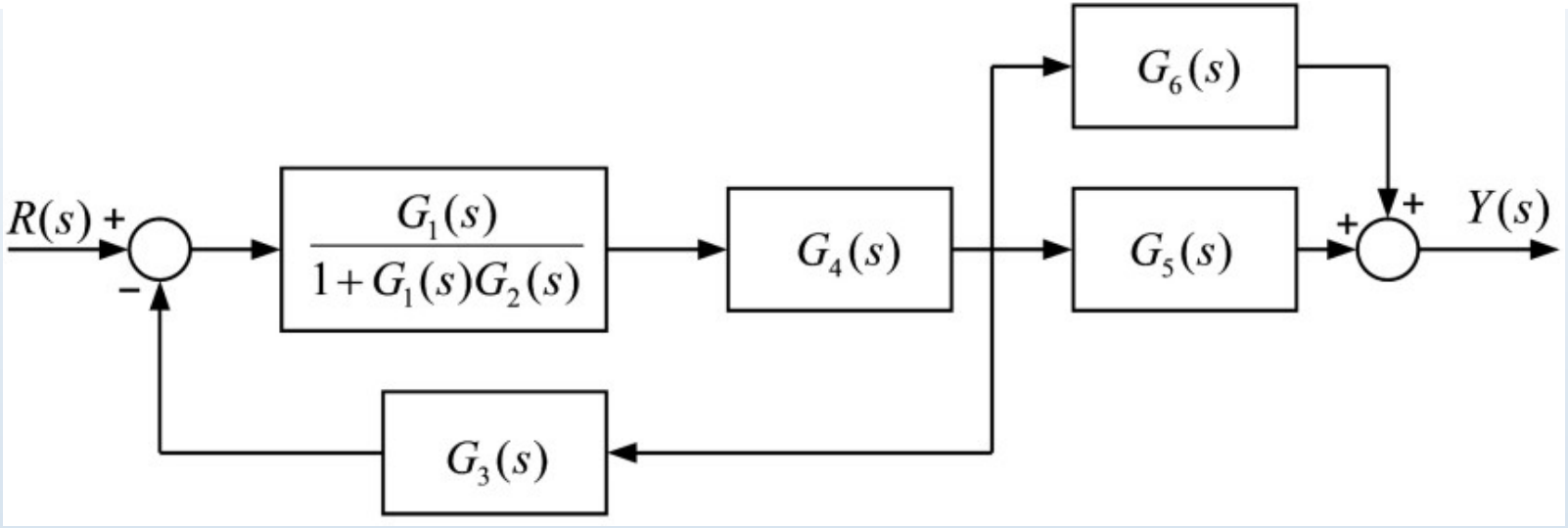
Block Diagram Algebra

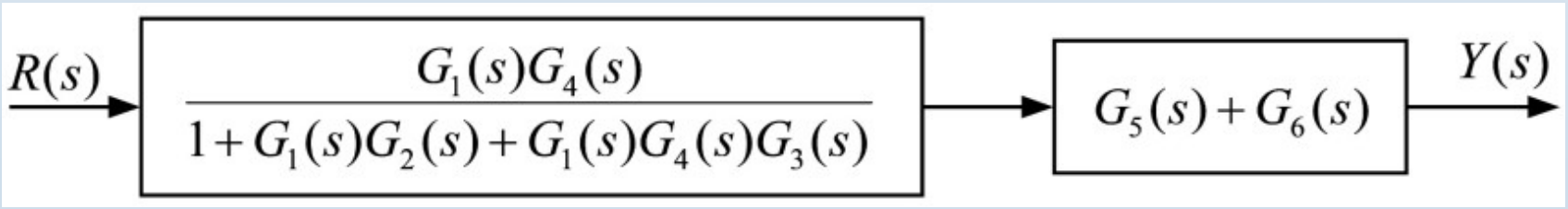


Block Diagram Algebra: Example 1



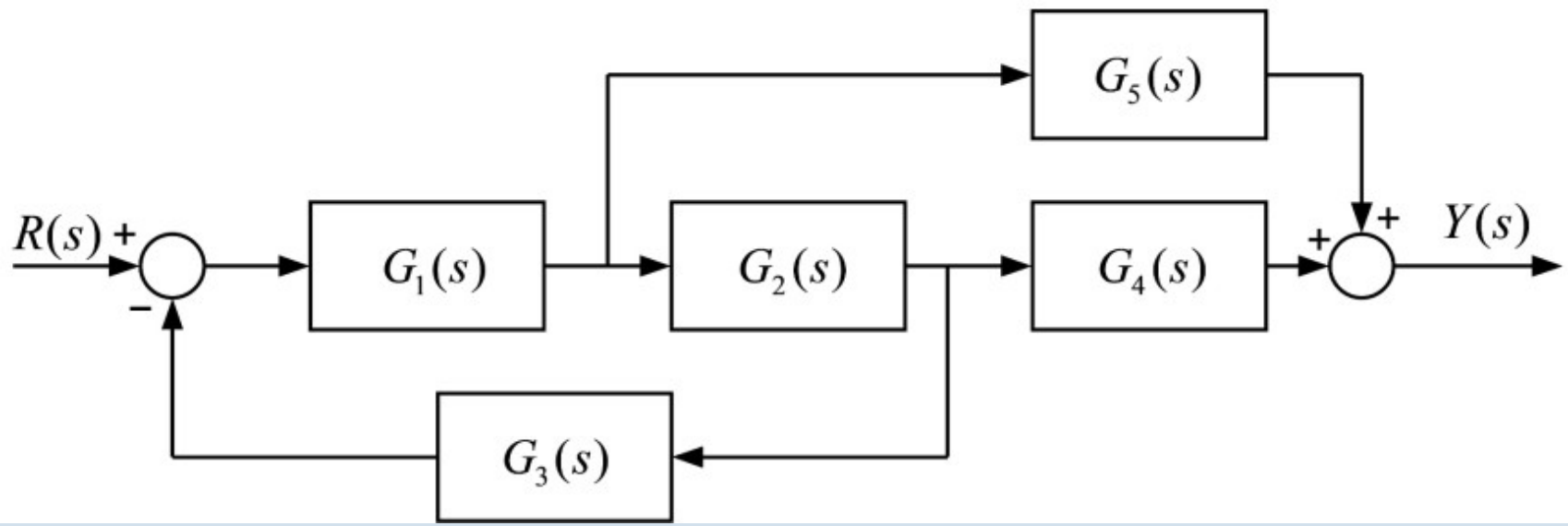
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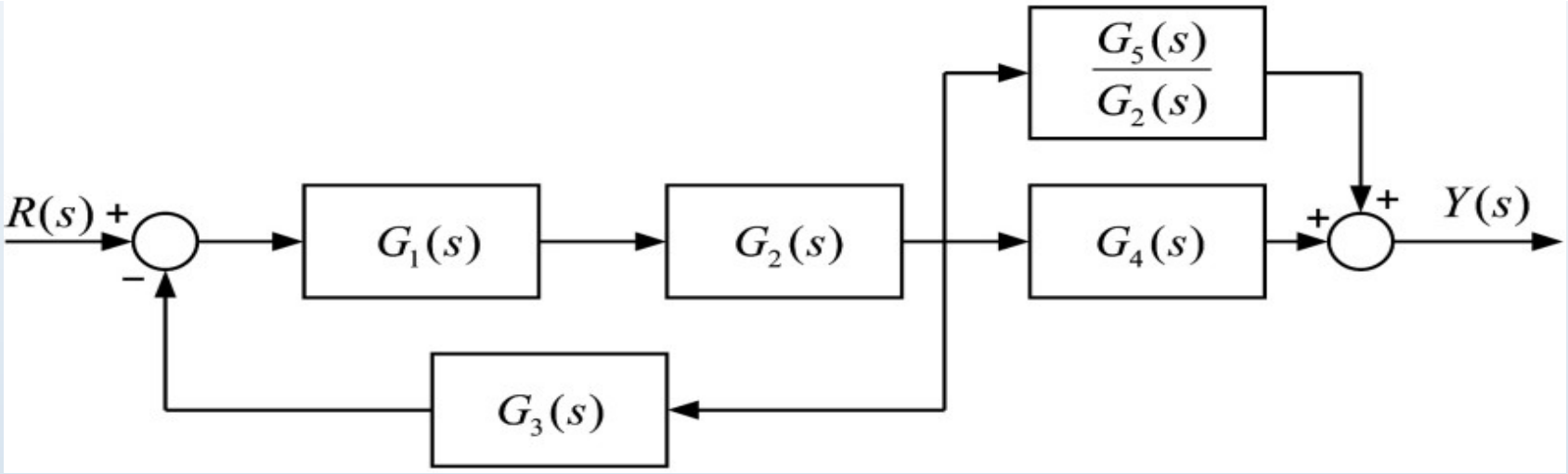


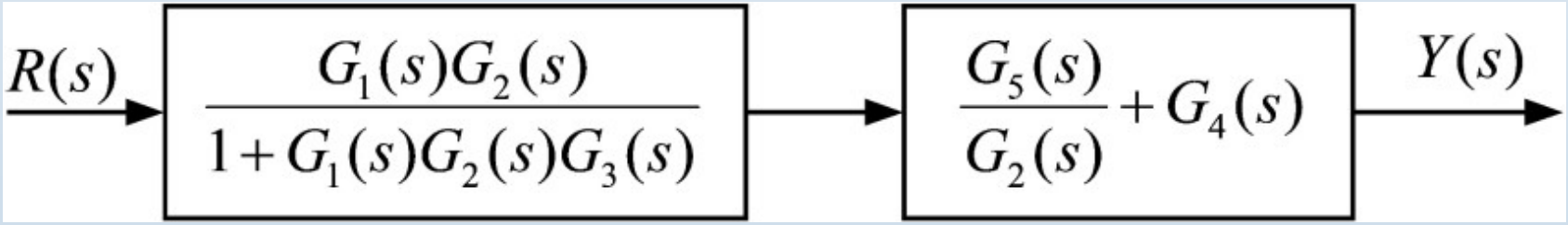
$$G(s) = \frac{Y(s)}{R(s)} = \frac{G_1(s)G_4(s)(G_5(s)+G_6(s))}{1+G_1(s)G_2(s)+G_1(s)G_4(s)G_3(s)}$$

Block Diagram Algebra: Example 2



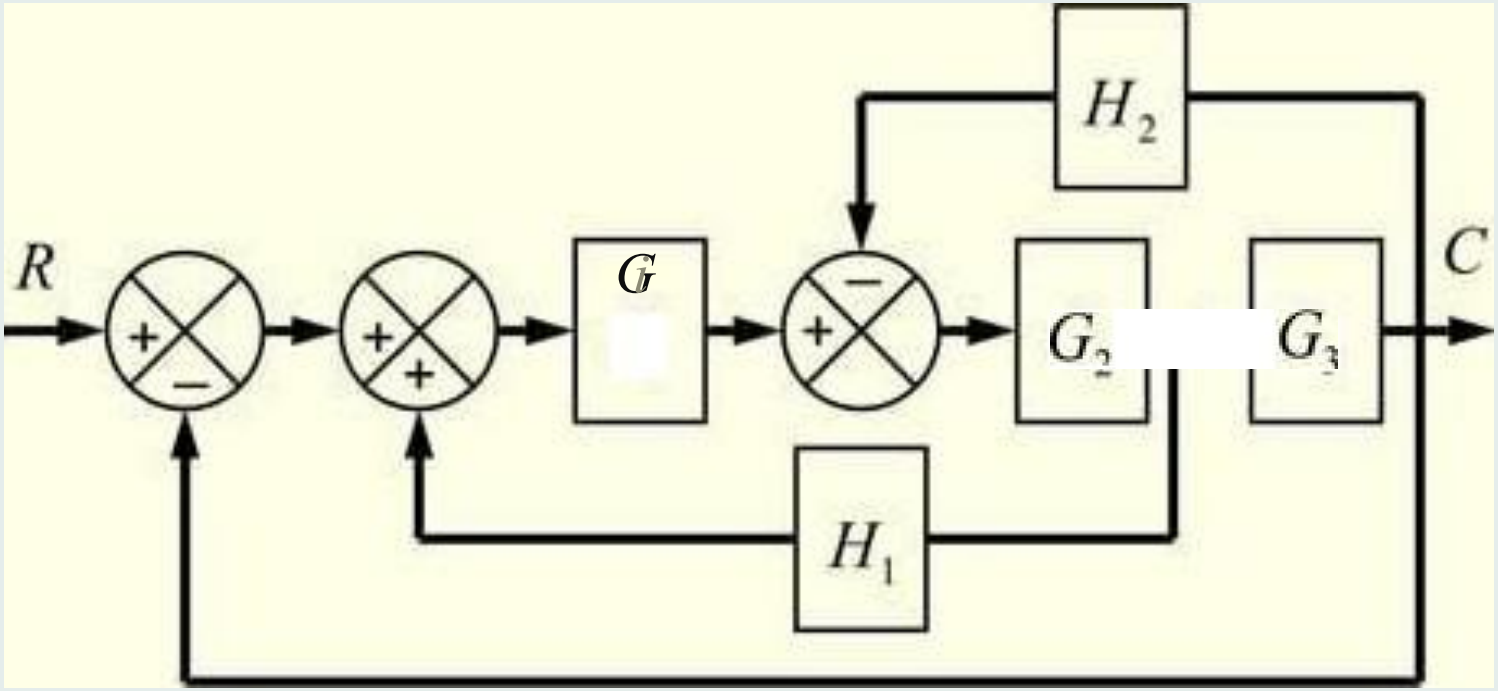
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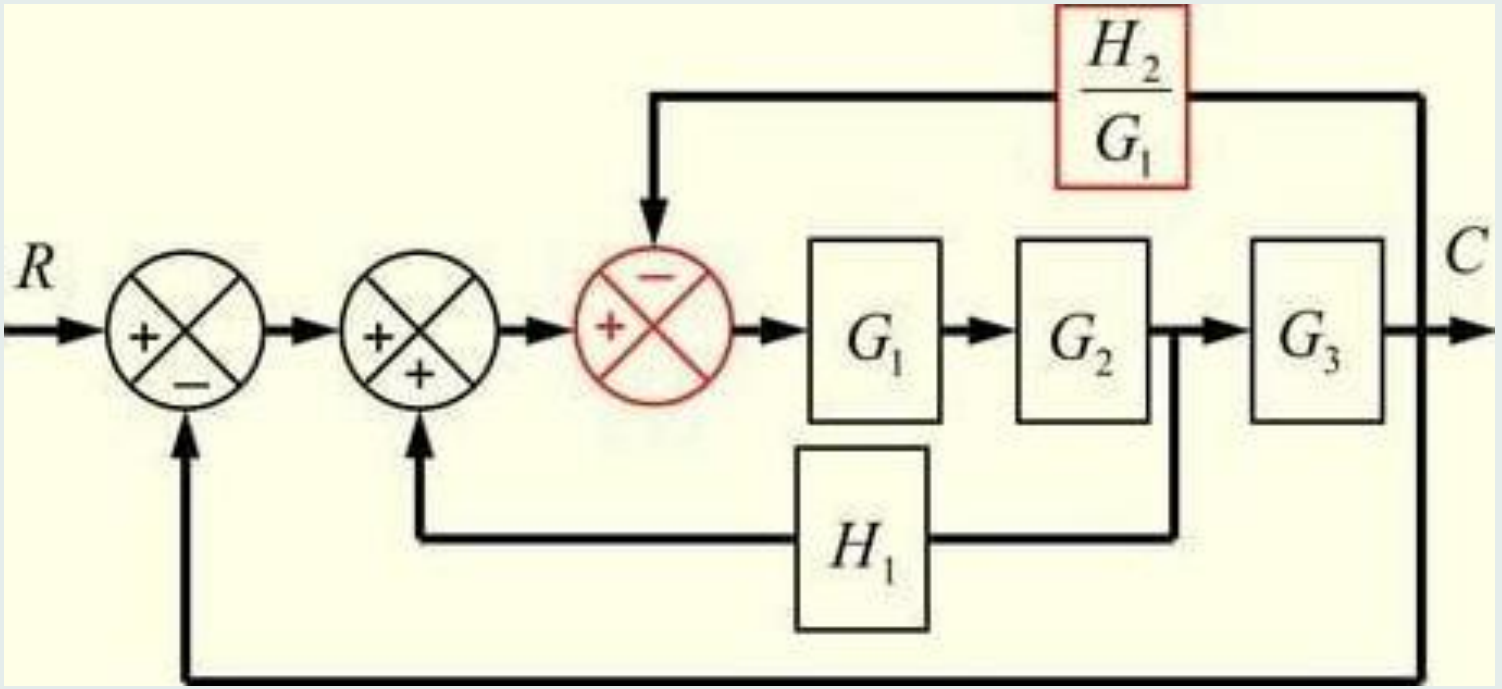


$$G(s) = \frac{Y(s)}{R(s)} = \frac{G_1(s)G_5(s) + G_1(s)G_2(s)G_4(s)}{1 + G_1(s)G_2(s)G_3(s)}$$

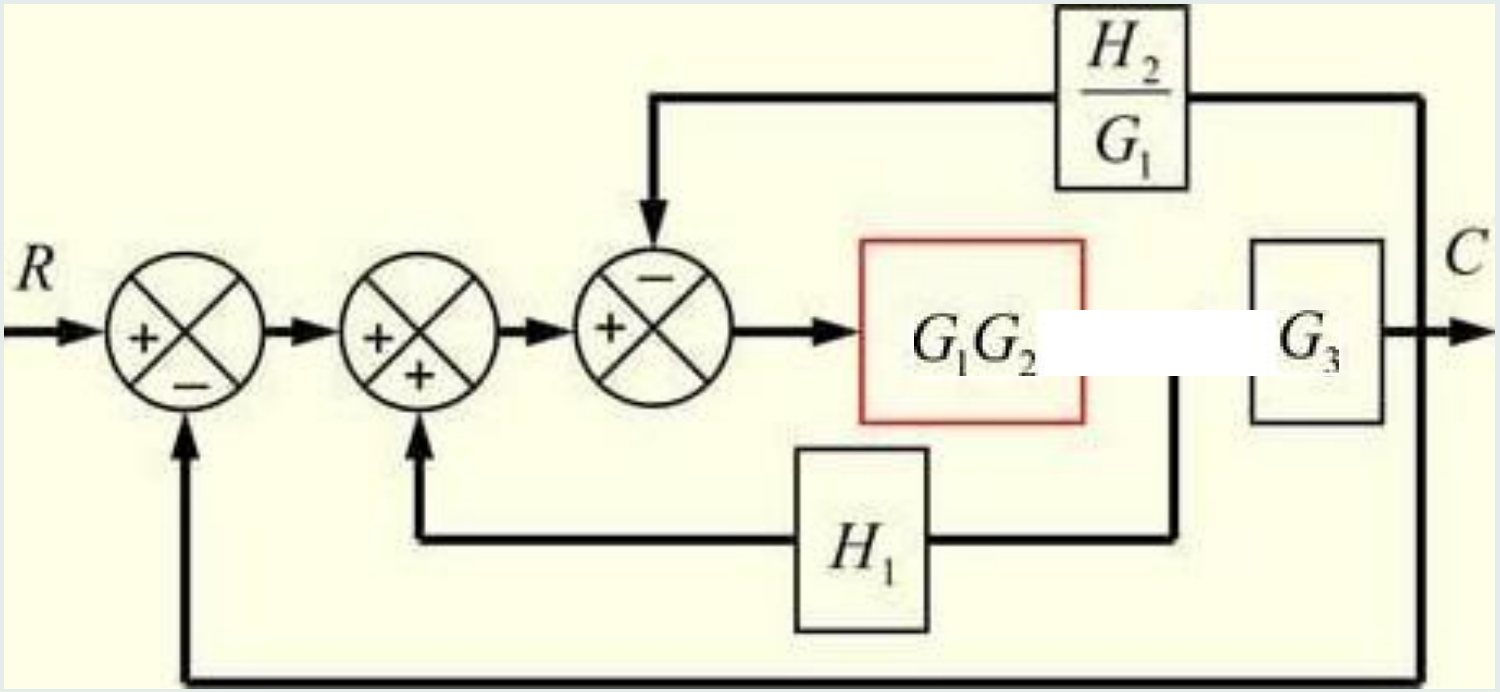
block diagram: Example # 3



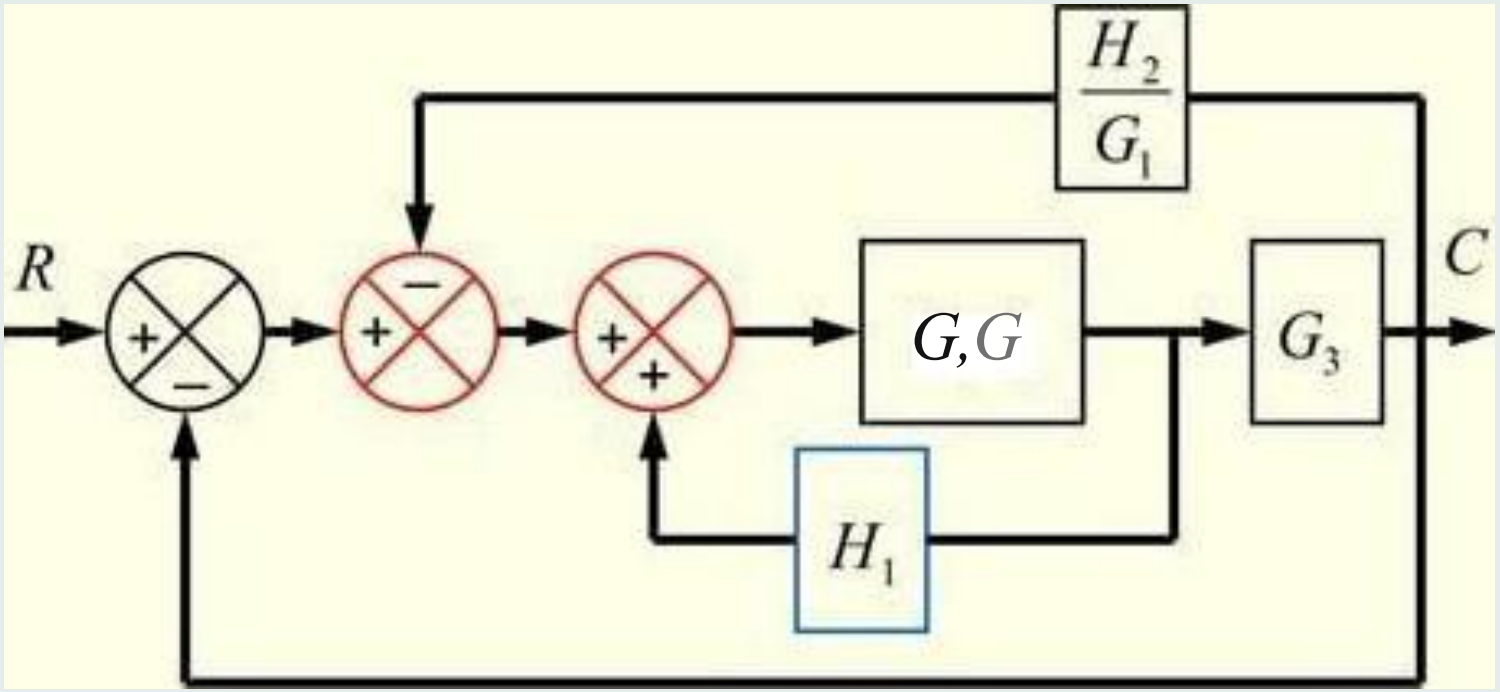
Moving the summing point ahead of Gi,



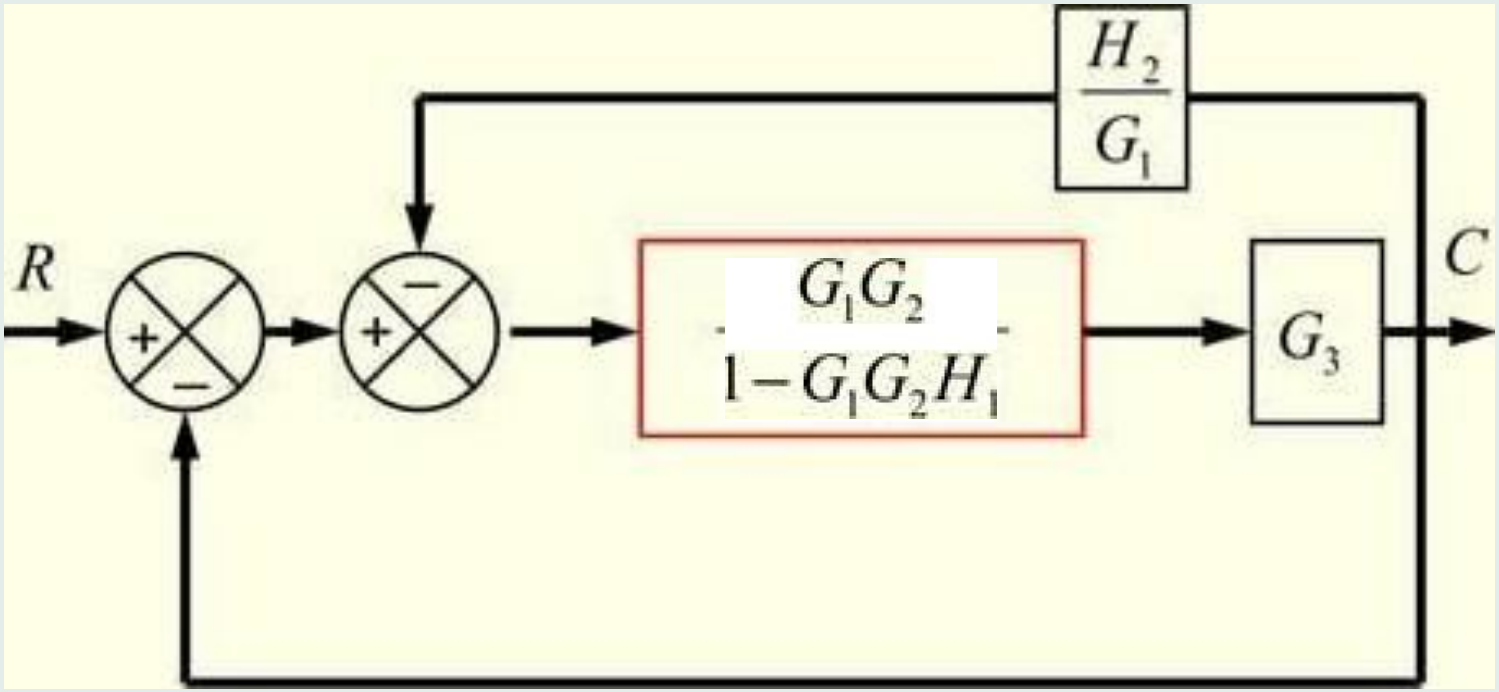
G1 and G2 are in series



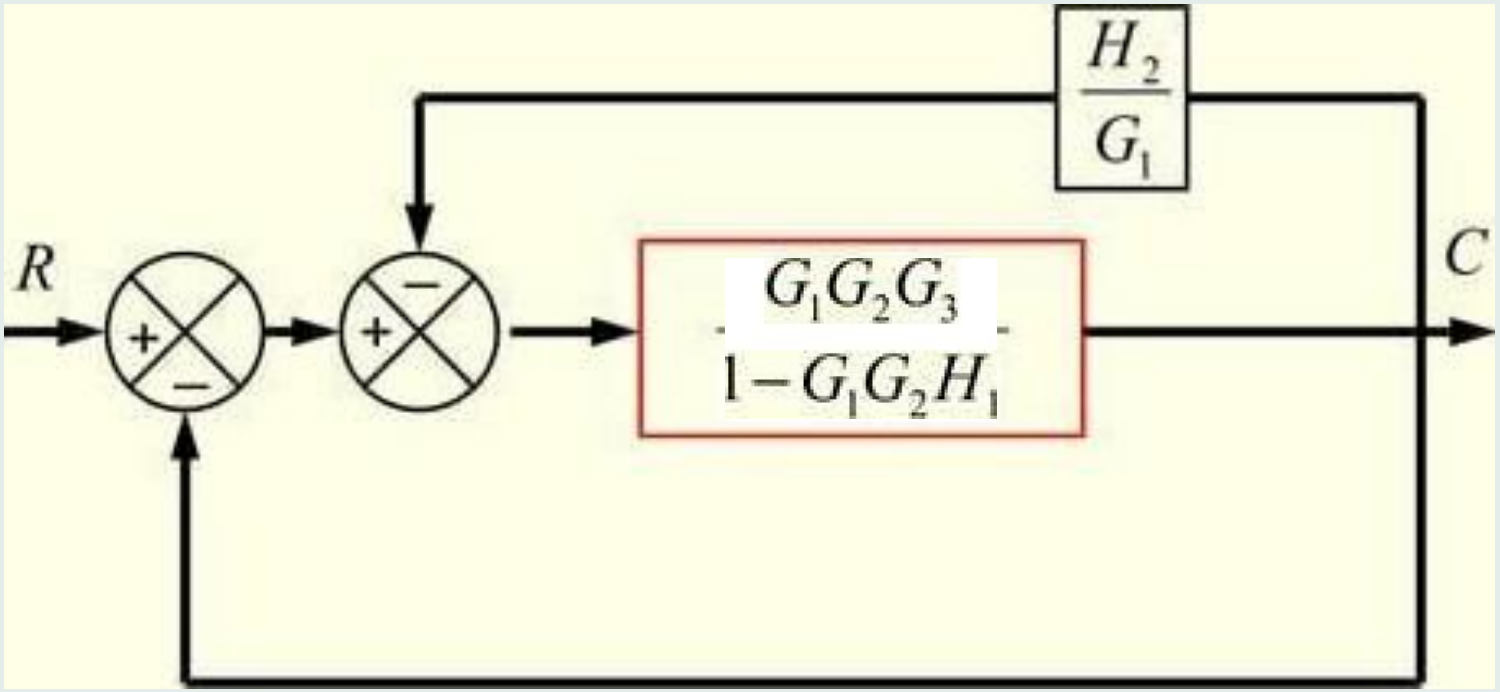
feedback loop G1, G2 and H1



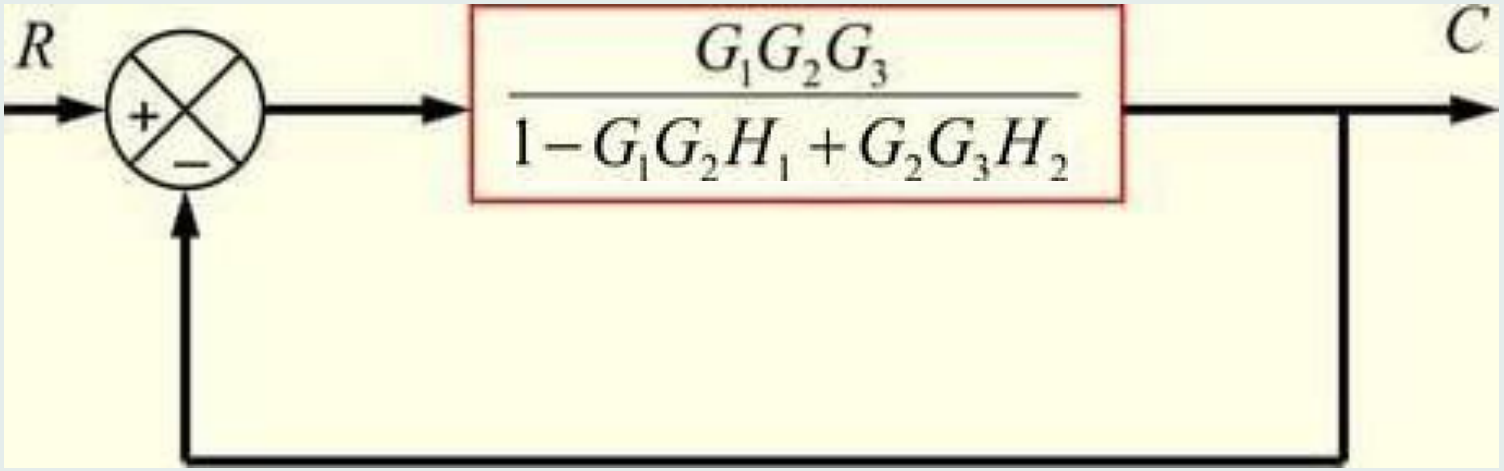
two blocks in Series



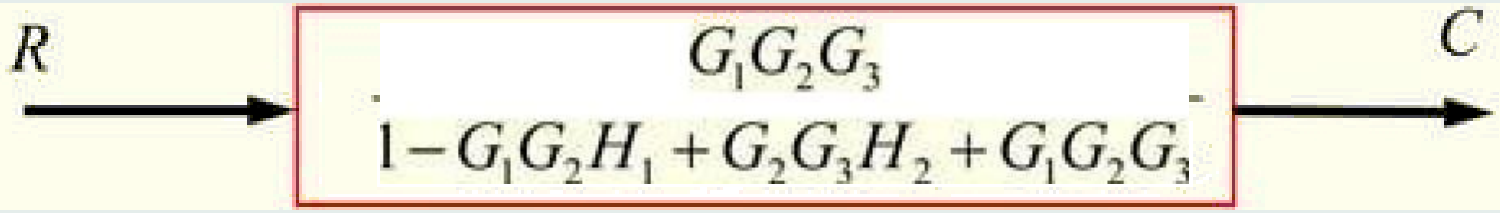
second feedback loop we get:



third feedback loop we get:

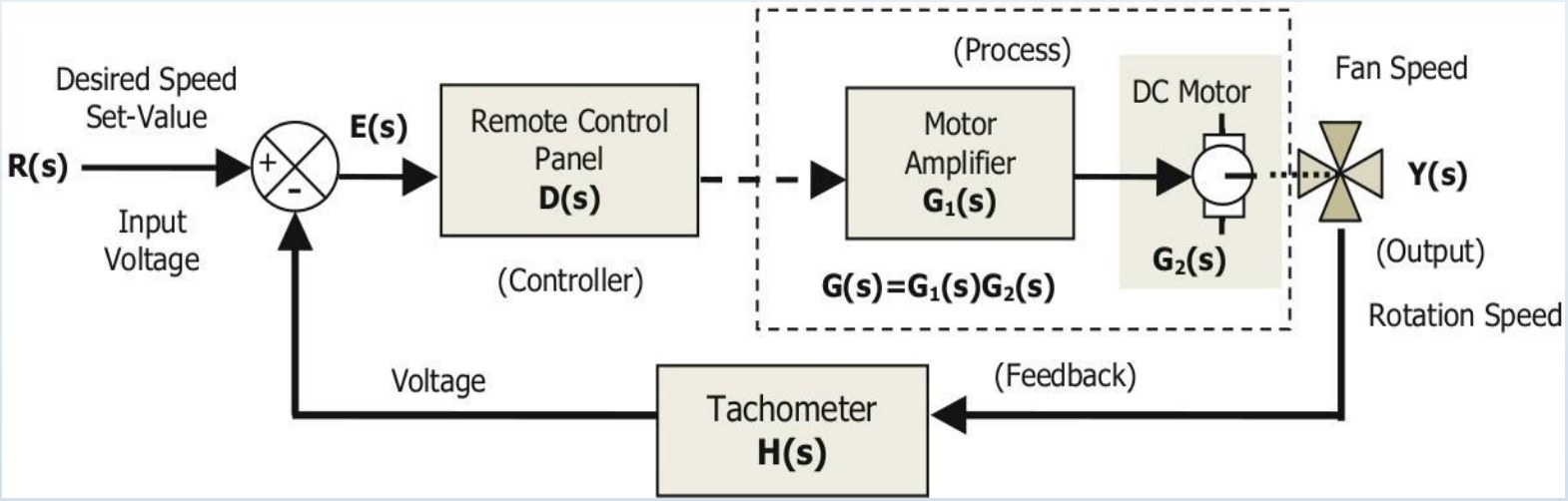


The system is reduced to the following block diagram:



Transfer Function Example of DC Motor

What is the Transfer functions of this system?

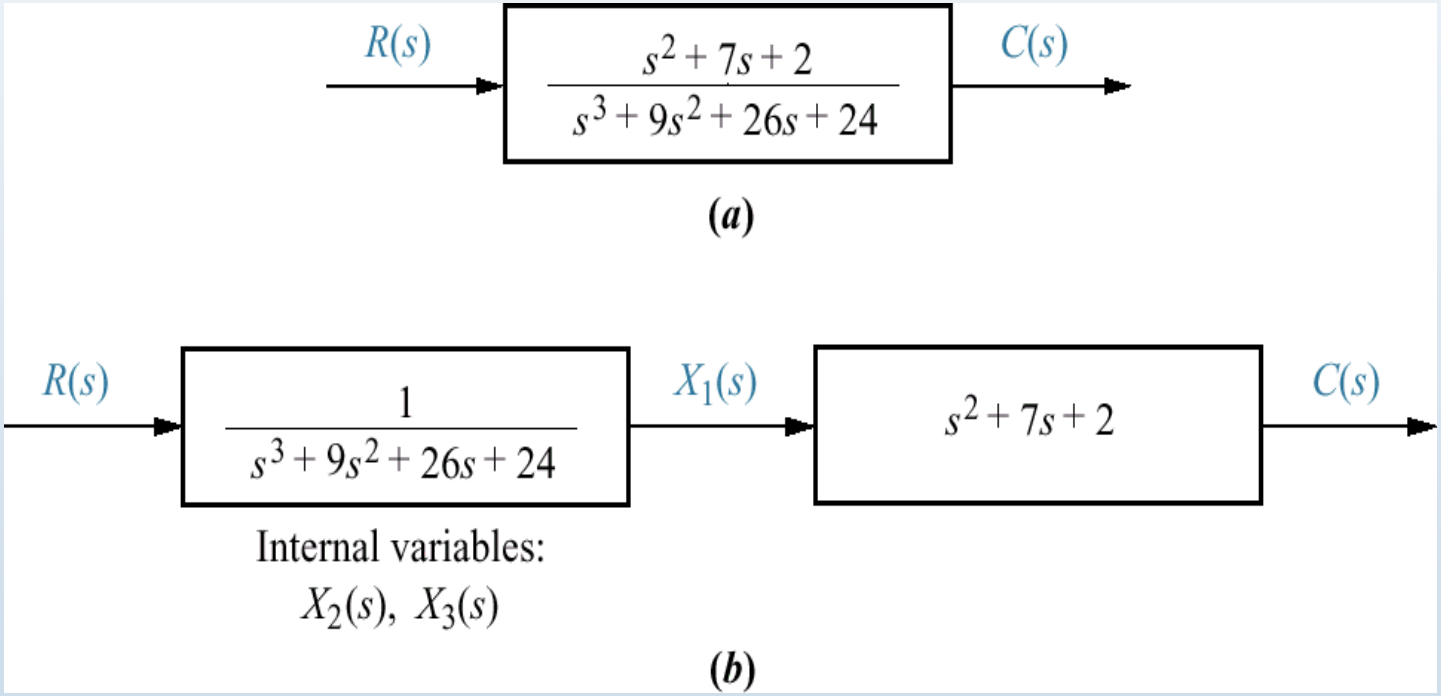


$$Y(s) = D(s)G(s)E(s)$$

$$E(s) = R(s) - H(s)Y(s)$$

$$\frac{Y(s)}{R(s)} = \frac{D(s)G(s)}{1 + D(s)G(s)H(s)}$$

Decomposed Transfer Function



Laplace Transform

- Transform complicated DEs to relatively simple algebraic equations

$$F(s) = \int_{0^-}^{\infty} f(t)\mathrm{e}^{-st}\mathrm{d}t = L\{f(t)\}$$

$$s \equiv \frac{\mathrm{d}}{\mathrm{d}t}$$

$$\frac{1}{s} \equiv \int_{0^-}^{t} \mathrm{d}t$$

Laplace Transform Table

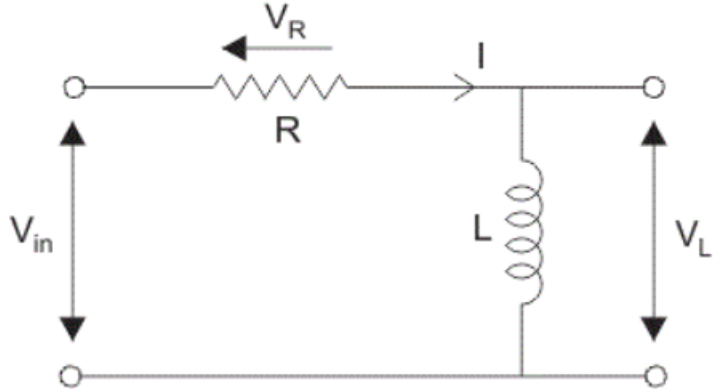
$f(t)$	$F(s)$
Step function $u(t)$	$\frac{1}{s}$
e^{-at}	$\frac{1}{s+a}$
$\sin\ \omega t$	$\frac{\omega}{s^2+\omega^2}$
$\cos\ \omega t$	$\frac{s}{s^2+\omega^2}$
$e^{-at}f(t)$	$F(s+a)$
t^n	$\frac{n!}{s^{n+1}}$
$f^{(k)}(t)=\frac{\mathrm{d}^k f(t)}{\mathrm{d}t^k}$	$s^kF(s)-s^{k-1}f(0^-)-s^{k-2}f'(0^-)-\cdots-f^{(k-1)}(0^-)$
$\int_{-\infty}^{t} f(t)\mathrm{d}t$	$\frac{F(s)}{s}+\frac{1}{s}\int_{-\infty}^{0} f(t)\mathrm{d}t$
Impulse function $\delta(t)$	1

Laplace Transform Examples

- Transfer functions for
  - RL Circuit
  - RC Circuit
  - RLC Circuit
  - DC motor

RL Circuit

- **RL Circuit**: An RL circuit is defined as a circuit that includes both a resistor and an inductor, either in series or parallel, connected to an input voltage.



- **Transfer Function of Series RL Circuit**

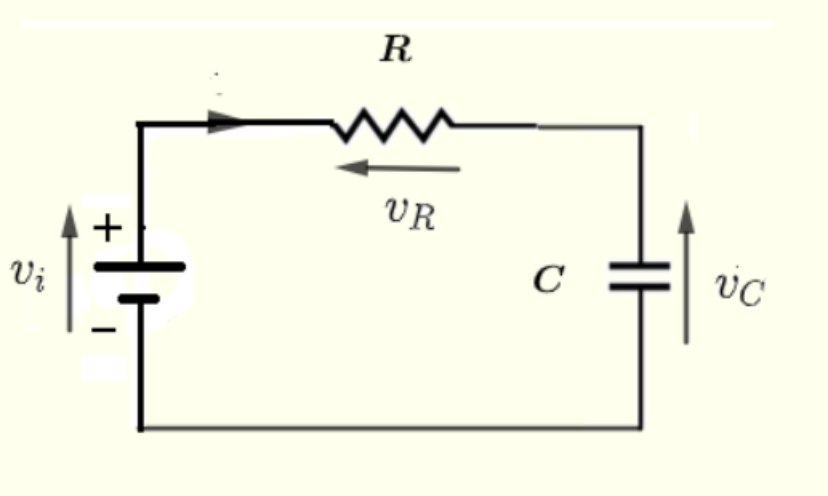
- Current in resistor and inductor are the same,

- T.F $$= \frac{V_L(S)}{V_{in}(s)} = \frac{LS}{(R + LS)}$$

RC Circuit

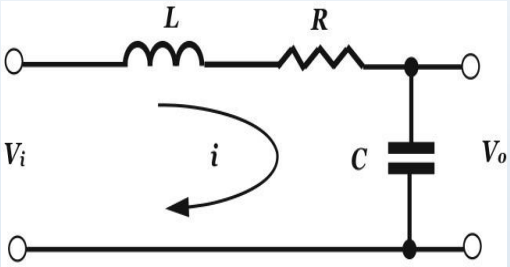
- **RC Circuit**: An RC circuit is defined as a circuit that includes a resistor and a capacitor connected to an input voltage.

- **Transfer Function of Series RC Circuit**



$$V_C(s) = \frac{1}{1+RCs} V_{in}(s)$$

RLC Circuit

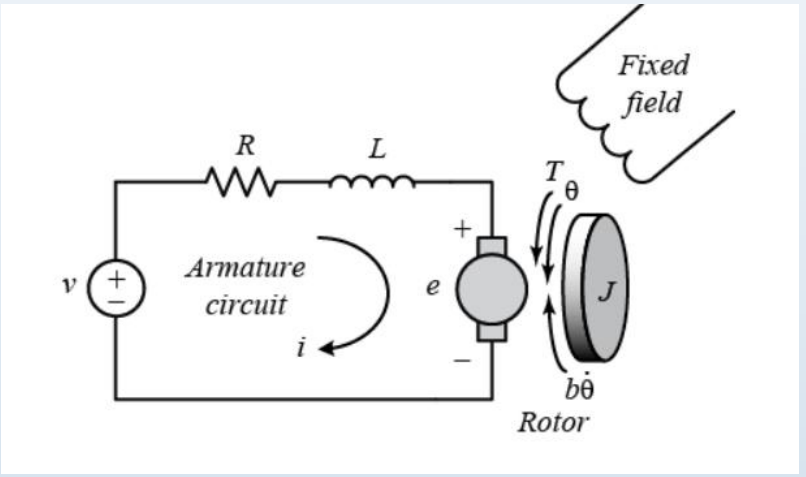


$$L\frac{\mathrm{d}i}{\mathrm{d}t} + Ri + \frac{1}{C}\int i\mathrm{d}t = V_i$$

$$G(s) = \frac{V_O(s)}{V_i(s)} = \frac{1}{LCs^2 + RCs + 1}$$

DC Motor Speed: System Modeling

- DC motor provides rotary motion to the wheels of a robot.
- Model/Equation
- The torque generated by a DC motor is proportional to the armature current
- The back emf, e, is proportional to the angular velocity of the shaft
- Newton's 2nd law and Kirchhoff's voltage law.



$$e = K_e\dot{\theta}$$

$$T = K_t i$$

$$J\ddot{\theta} + b\dot{\theta} = Ki$$

$$L\frac{di}{dt} + Ri = V - K\dot{\theta}$$

- Taking Laplace TF

$$s(Js+b)\,\theta(s) = KI(s)$$

$$(Ls+R)I(s) = V_i(s) - Ks\theta(s)$$

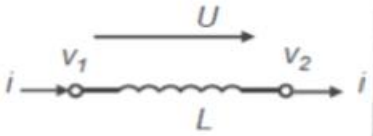
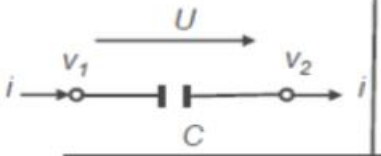
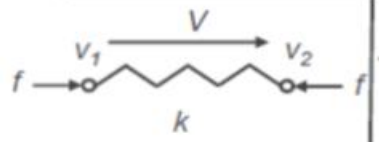
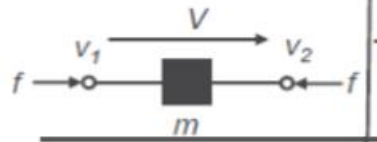
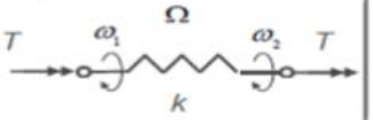
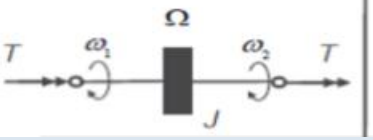
Combining Eqs. (4.13) and (4.14) to eliminate the intermediate variable I(s), we get the transfer function for this DC motor system as

$$G(s) = \frac{\Omega(s)}{V_i(s)} = \frac{K}{(Js+b)(Ls+R)+K^2} \quad \left(\frac{\text{rad/s}}{\text{V}}\right) \tag{4.15}$$

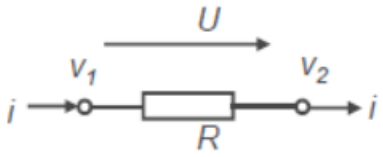
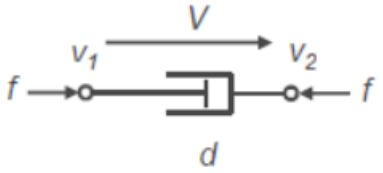
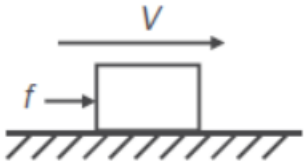
where $\dot{\theta}(s) = s\theta(s) = \Omega(s)$ and it is the rotation speed of the motor (ω) measured by rad/s.

This is a second-order system with two poles and no zeros, and it is a typical transfer function used for most linear systems.

Basic Elements for modeling

Symbol	Physical element	Constitutive relation	Stored energy
	Inductance	$L\frac{di}{dt}=U=v_1-v_2$	$E=\frac{1}{2}Li^2$
	Capacitance	$C\frac{dU}{dt}=i, U=v_1-v_2$	$E=\frac{1}{2}Cu^2$
	Translational Spring	$f=ky, y=\int V dt$ $\frac{1}{k}\frac{df}{dt}=V$	$E=\frac{1}{2}ky^2$
	Translational Mass	$m\frac{dV}{dt}=f, V=v_1=v_2$	$E=\frac{1}{2}mV^2$
	Rotational Spring	$T=k\varphi, \varphi=\int \Omega dt$ $\dot{T}=k\phi$	$E=\frac{1}{2}k\varphi^2$
	Rotational mass	$J\frac{d\Omega}{dt}=T, \Omega=\omega_1=\omega_2$	$E=\frac{1}{2}m\Omega^2$

Continue

Symbol	Physical element	Constitutive equation	Stored energy
	Resistance	$U = Ri$	———
	Damping (friction)	$f = dV$	———
	Dry friction	$f = \mu mg$	———

Laws & Rules in modeling

- Mechanical properties -> Newtons laws
- Electrical properties -> Kirchhofs laws
- Parallel and series equations
- The order of the differential equations equals the number of energy storage elements

Assignment#2

Questions